INTRODUCTION TO APPLIED NUMERICAL METHODS

The Newton Raphson Method for Roots of Equations

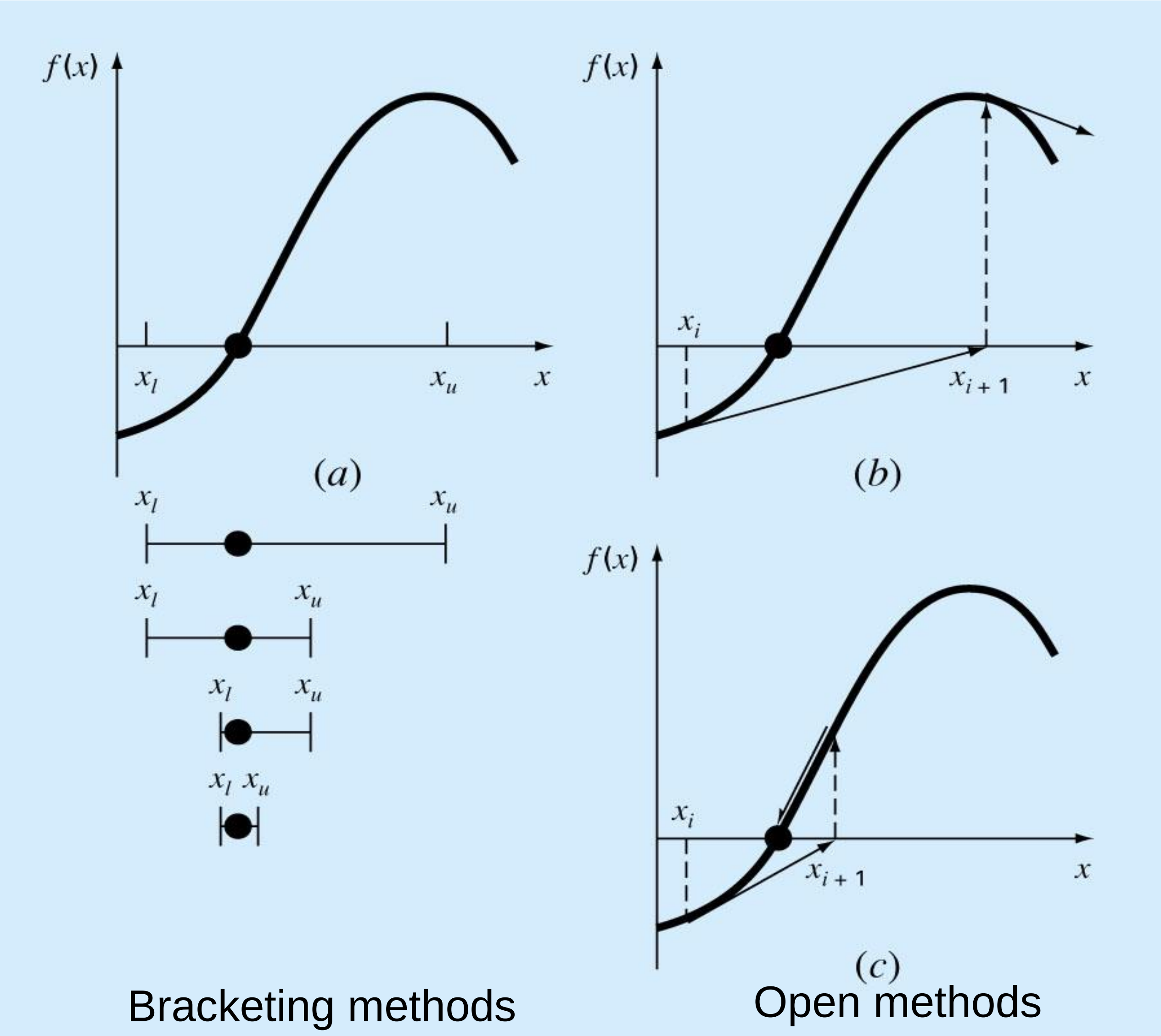
Lang Yuan, Associate Professor

UNIVERSITY OF
South Carolina

College of Engineering
and Computing

OPEN METHODS

- Open methods are based on formulas that require only a single starting value of x or two starting values that do not necessarily bracket the root.
- It can *diverge* from the true root



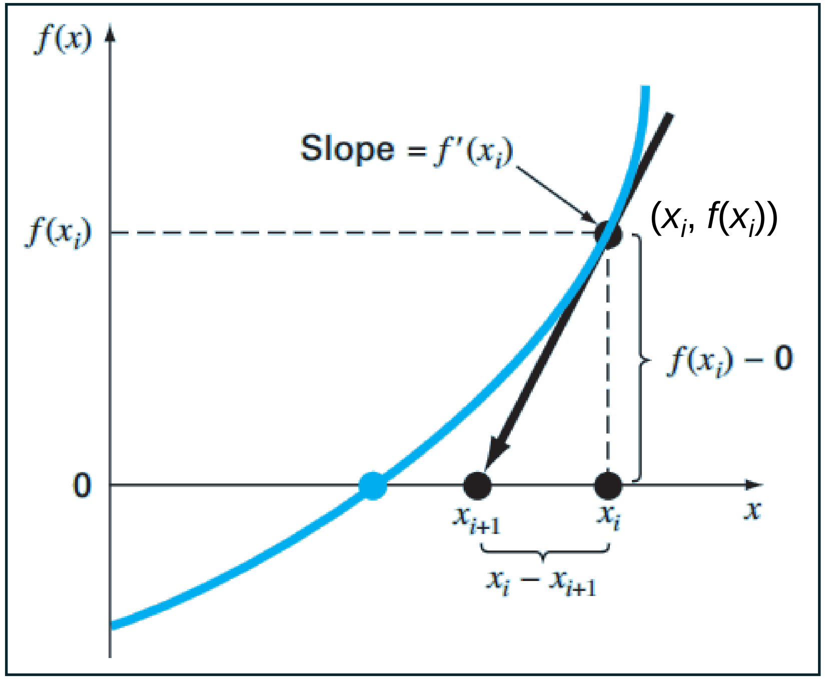
THE NEWTON-RAPHSON METHOD

Principle

- Open method
- Most widely used root-finding method
- Use tangent to obtain an improved estimate for the true root

Approach 1: Geometrical Interpretation

- A tangent $f'(x_i)$ is drawn from $(x_i, f(x_i))$
- **The point where the tangent crosses the x-axis is an improved estimate of the root**



Numerical Methods for Engineers, 7th Ed.
(S.C.Chapra and R.P.Canale)

Approximation of the First-order Derivative using Forward Difference

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}} \Rightarrow \boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

Newton-Raphson Method

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THE NEWTON-RAPHSON METHOD

Approach 2: Taylor Series

- Estimate $f(x_{i+1})$ @ the expansion point x_i
- Truncation the terms of the 2nd order term and the higher-order terms

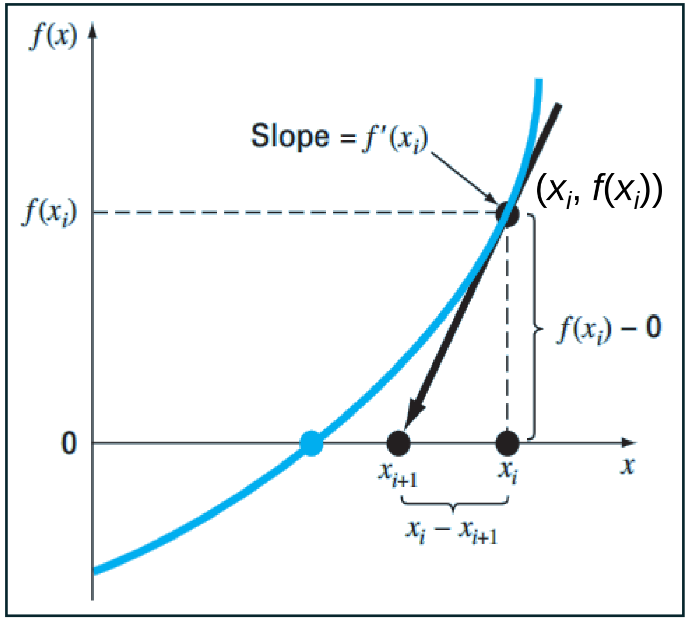
$$\boxed{f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)}$$

intersection with the x-axis: $f(x_{i+1}) = 0$

$$\boxed{0 = f(x_i) + f'(x_i)(x_{i+1} - x_i)}$$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

Expression available for use



1 Guess/Find x_i on x-axis → 2 Find $f(x_i)$ on the curve f(x) → 3 Draw a tangent @ $(x_i, f(x_i))$

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AN EXAMPLE

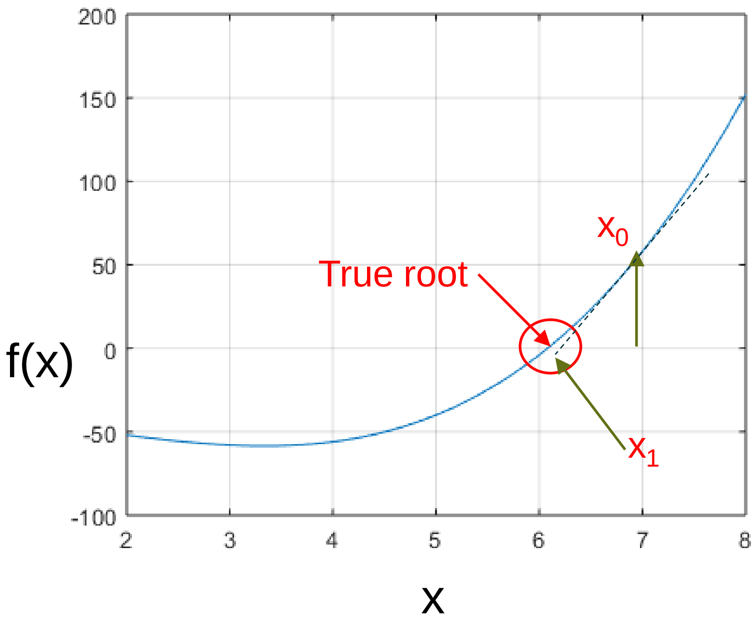
Use the Newton-Raphson method to locate the root of

$$f(x) = x^3 - 5x^2 - 40 \qquad \text{using an initial guess of } x_0 = 7.0$$

Solution:

The first derivative of f(x) is $f'(x) = 3x^2 - 10x$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}} \Rightarrow \boxed{x_{i+1} = x_i - \frac{x_i^3 - 5x_i^2 - 40}{3x_i^2 - 10x_i}}$$



Step 1: Substitute $x_0 = 7.0$ to obtain x_1

$$x_1 = x_0 - \frac{x_0^3 - 5x_0^2 - 40}{3x_0^2 - 10x_0} = 7.0 - \frac{7^3 - 5\times 7^2 - 40}{3\times 7^2 - 10\times 7} = 7.0 - \frac{98 - 40}{77} = 6.246$$

Step 2: Substitute $x_1 = 6.246$ to obtain x_2

$$x_2 = x_1 - \frac{x_1^3 - 5x_1^2 - 40}{3x_1^2 - 10x_1} = 6.246 - \frac{6.246^3 - 5\times 6.246^2 - 40}{3\times 6.246^2 - 10\times 6.246} = 6.246 - \frac{8.6096}{54.5775} = 6.088$$

AN EXAMPLE

Use the Newton-Raphson method to locate the root of

$f(x) = x^3 - 5x^2 - 40$ using an initial guess of $x_0 = 7.0$

Solution:

The first derivative of f(x) is $f'(x) = 3x^2 - 10x$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}} \Rightarrow \boxed{x_{i+1} = x_i - \frac{x_i^3 - 5x_i^2 - 40}{3x_i^2 - 10x_i}}$$

Step 3: Substitute $x_2 = 6.088$ to obtain x_3

$$x_3 = x_2 - \frac{x_2^3 - 5x_2^2 - 40}{3x_2^2 - 10x_2} = 6.088 - \frac{6.088^3 - 5\times 6.088^2 - 40}{3\times 6.088^2 - 10\times 6.088} = 6.246 - \frac{8.6096}{54.5775} = 6.0815$$

Approximate percent relative error

present iteration previous iteration

$$\varepsilon_a = \left|\frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}}\right| 100\% = \left|\frac{6.0815 - 6.088}{6.0815}\right| \times 100\% = 0.11$$

ANOTHER EXAMPLE

Use the Newton-Raphson method to locate the root of

$f(x) = e^{-x} - x$ using an initial guess of $x_0 = 0$

Solution:

The first derivative of f(x) is $f'(x) = \frac{de^{-x}}{dx} - 1 = e^{-x}\frac{d(-x)}{dx} - 1 = -e^{-x} - 1$

How to compute $\frac{de^{-x}}{dx}$?

Chain rule: $f(x) = F(g(x)) \Rightarrow \frac{df(x)}{dx} = \frac{df}{dg}\frac{dg}{dx}$

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ANOTHER EXAMPLE

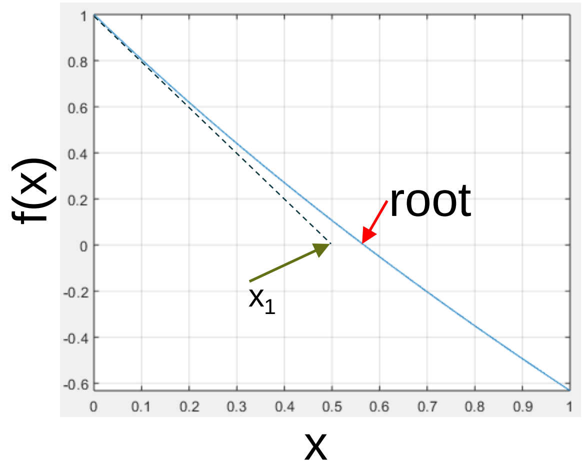
Use the Newton-Raphson method to locate the root of

$f(x) = e^{-x} - x$ using an initial guess of $x_0 = 0$

Solution:

The first derivative of f(x) is $f'(x) = \frac{\partial e^{-x}}{\partial x} - 1 = e^{-x}\frac{\partial(-x)}{\partial x} - 1 = -e^{-x} - 1$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}} \Rightarrow \boxed{x_{i+1} = x_i - \frac{e^{-x_i} - x_i}{-e^{-x_i} - 1}}$$



Step 1: Substitute $x_0 = 0$ to obtain x_1

$$x_1 = 0 - \frac{e^0 - 0}{-e^0 - 1} = \frac{-1}{-2} = 0.5$$

Step 2: Substitute $x_1 = 0.5$ to obtain x_2

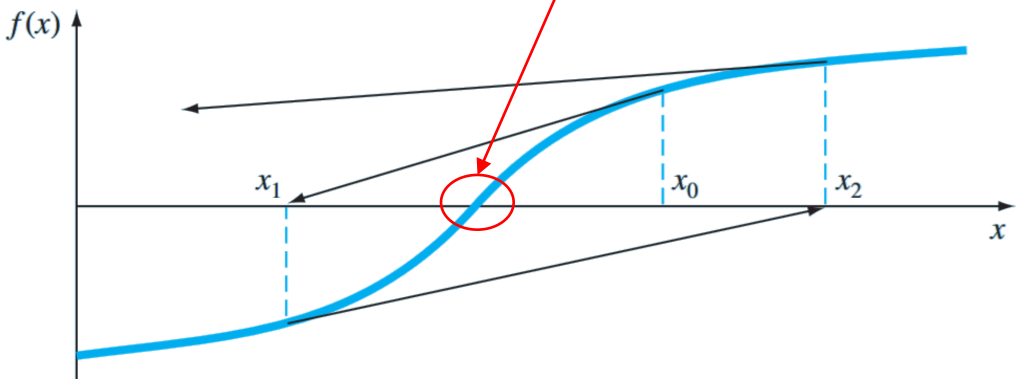
$$x_2 = 0.5 - \frac{e^{-0.5} - 0.5}{-e^{-0.5} - 1} = 0.5 - \frac{0.6065 - 0.5}{-0.6065 - 1} = 0.566311003$$

i	x_i	ε_t (%)
0	0	100
1	0.500000000	11.8
2	0.566311003	0.147
3	0.567143165	0.0000220
4	0.567143290	$< 10^{-8}$

DIFFICULTIES WITH NEWTON-RAPHSON METHOD

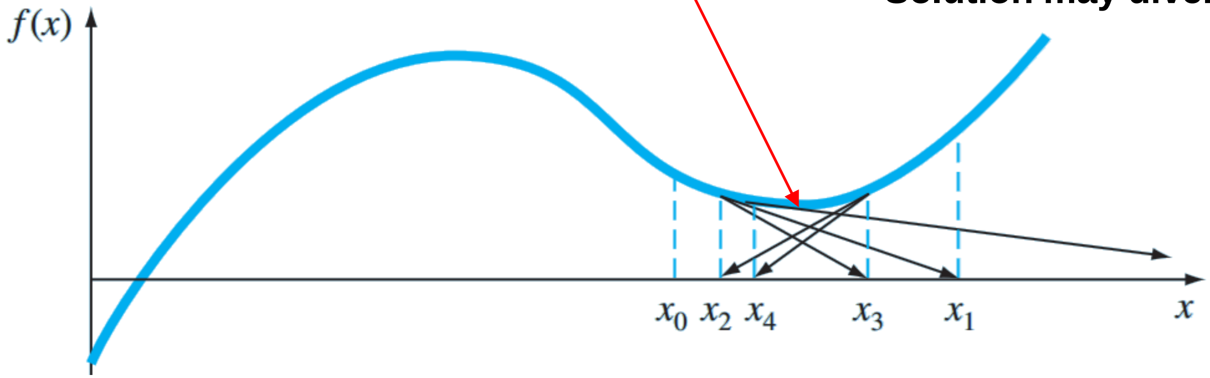
Case 1:

Inflection point f''(x) = 0 around the root ➡ Diverge from the root



Case 2:

A local minimum or maximum ➡ Oscillation around the minimum and maximum
Solution may diverge



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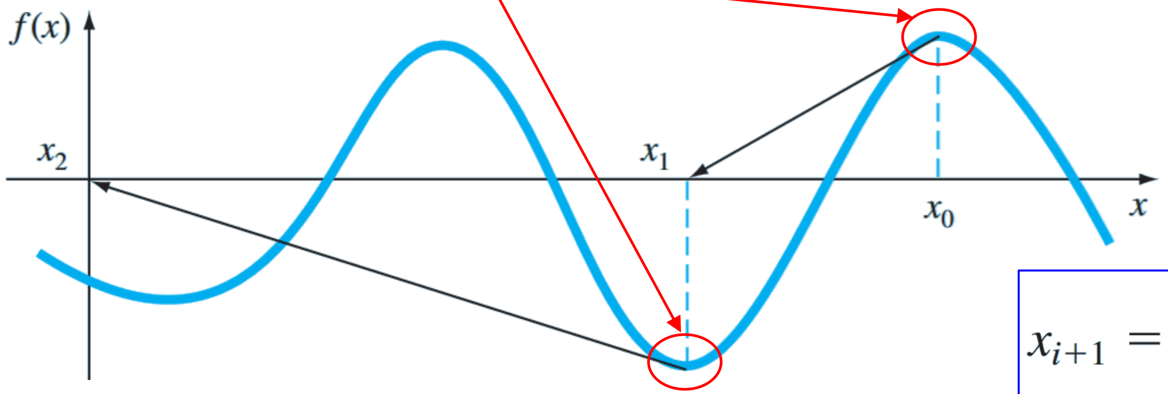
DIFFICULTIES WITH NEWTON-RAPHSON METHOD

Case 3:

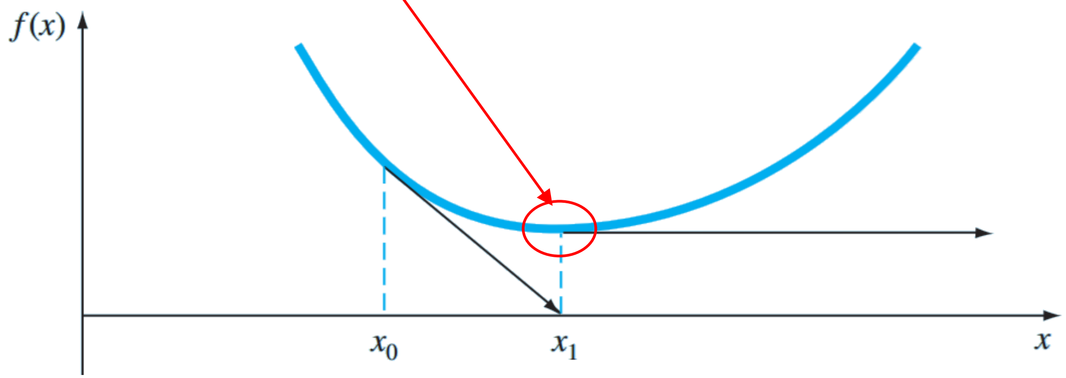
Near-zero slope f'(x) = 0 ➡ Jump away from the root (too large stride)



$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

Case 4:

Near-zero slope f'(x) = 0 ➡ Jump away from the root (too large stride)



DIFFICULTIES WITH NEWTON-RAPHSON METHOD

There is no general guidance to ensure convergence for the Newton-Raphson method

Its convergence depends on

- The nature of the function
- The accuracy of the initial guess: initial guess close to the root as possible (my trick: use Matlab to plot the function first to get a good initial guess and then to check the solution)

Potential improvements

- Substitute the solution back into the original function to check the residual (in order to avoid case 2)
- Include an upper limit on the number of iterations (to avoid oscillating, slowly convergent, or divergent solutions)
- Check if f'(x) approaches zero at any time during the computation